

Leveraging Allophony in Self-Supervised Speech Models for Atypical Pronunciation Assessment

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1. Background

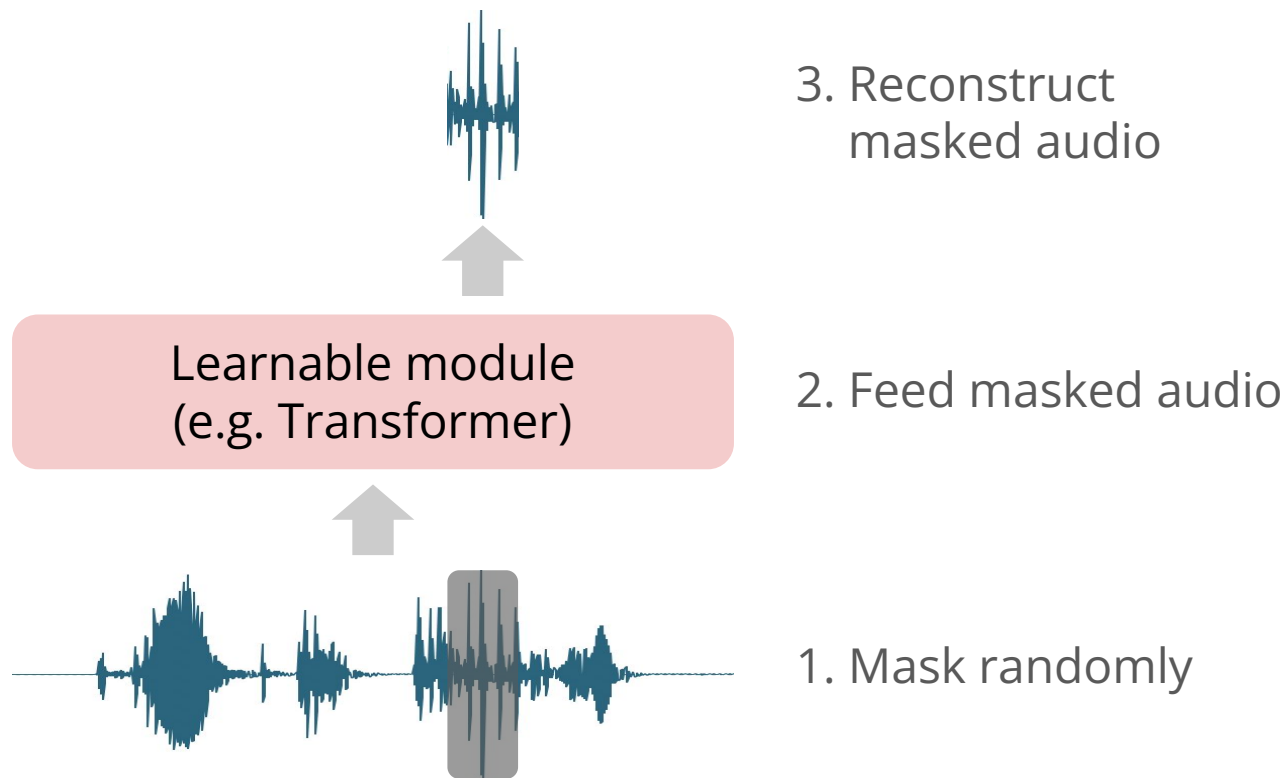
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Allophony

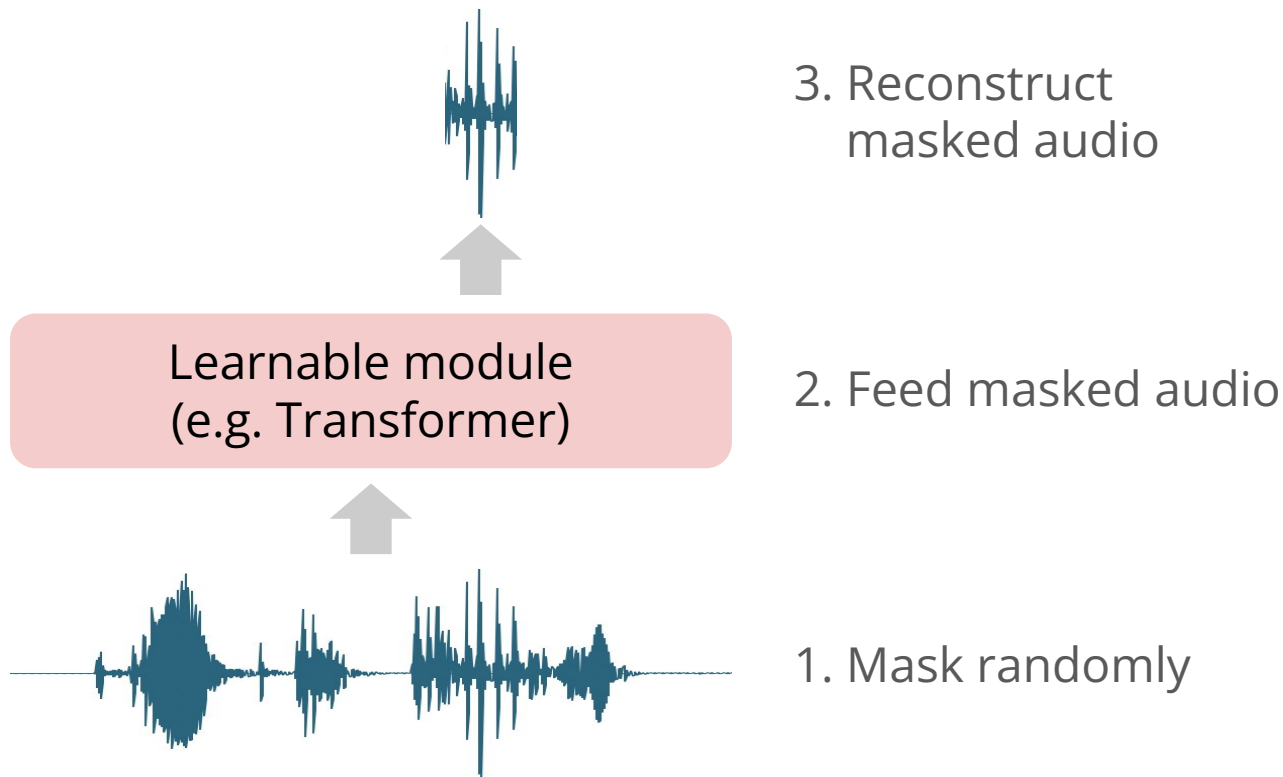
- A **phoneme** is a unit of sound that can make meaningful distinctions in a specific language.
 - We distinguish /pæt/ “pat” and /kæt/ “cat” because /p/ and /k/ is a different phoneme.
- **Allophones** are contextual realizations of phonemes.
 - Example. English phoneme /t/ is realized as multiple allophones.
 - [t^h] (aspirated stop) in “tap”
 - [t] (unaspirated stop) in “stop”
 - [ɾ] (flap) in “butter”
 - [ʔ] (glottal stop) in “kitten”

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Self-supervised Speech Models

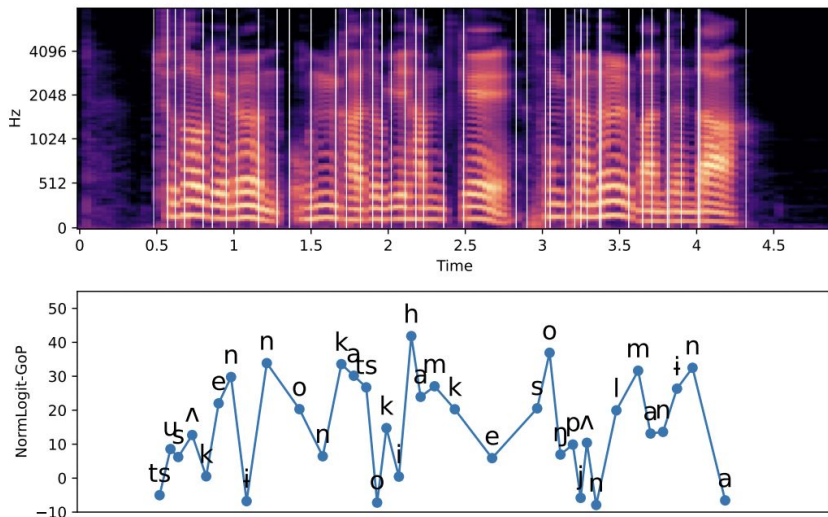


Self-supervised Speech Models



Leveraging **Allophony** in **Self-Supervised Speech Models** for **Atypical Pronunciation Assessment**

Atypical Pronunciation Assessment

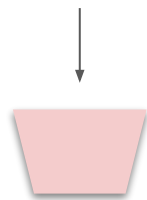
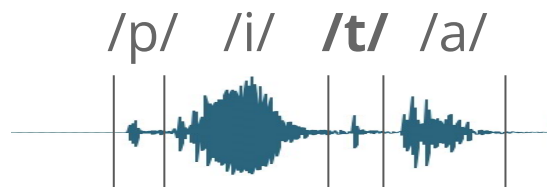


Goodness of Pronunciation (GoP) is conventionally used for atypical pronunciation assessment, such as pathological or non-native speech.

Phoneme-level pronunciation scores can detect mispronounced phonemes, making them useful for clinical or educational purposes.

Ref. Speech Intelligibility Assessment of Dysarthric Speech by using Goodness of Pronunciation with Uncertainty Quantification (Interspeech 2023)

Goodness of Pronunciation



$$P(/t/ \mid x) = \mathbf{0.8}$$

Textgrids

Phoneme + Start/End time

Phoneme Classifier

Trained with typical speech

GoP Score

Measures how typical the segment /i/ is

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2. Our method: MixGoP

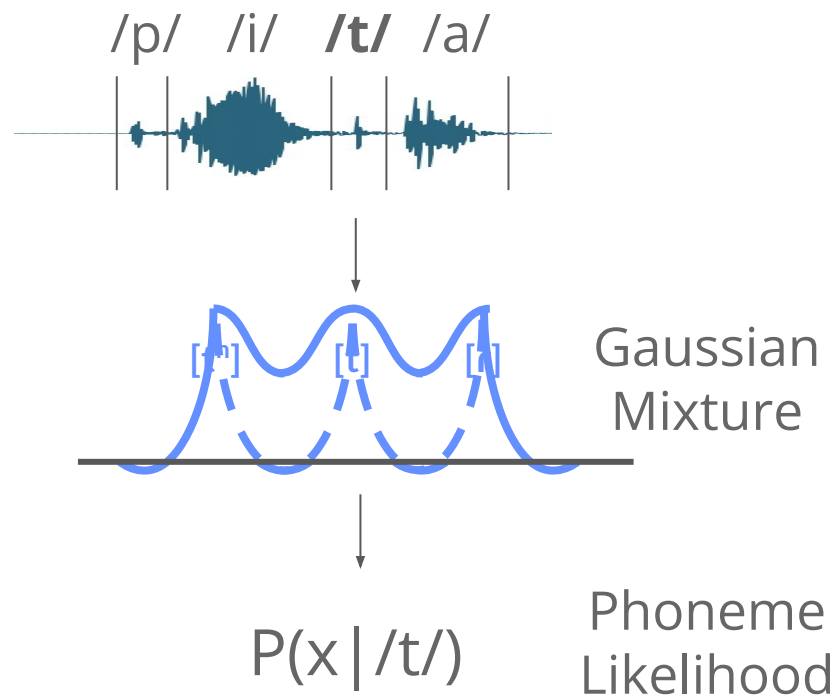
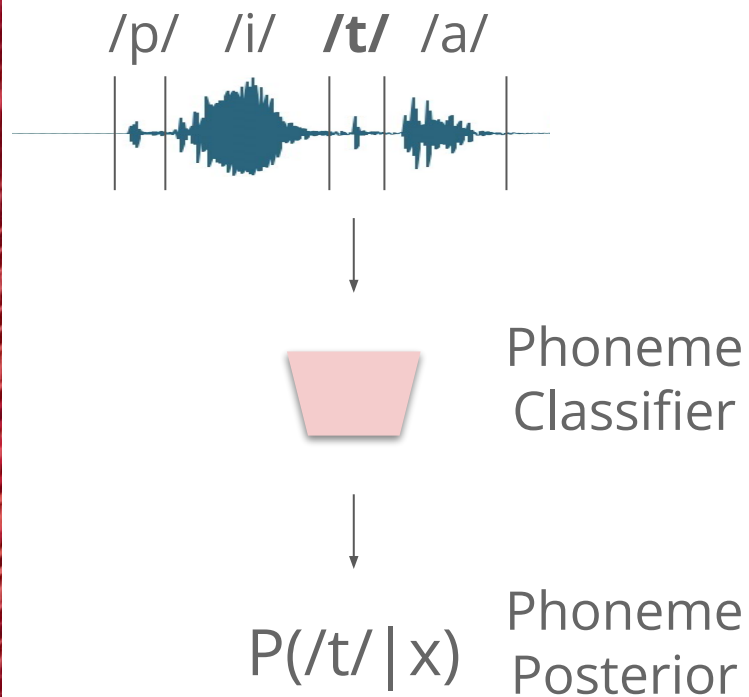
Limitations of Existing GoP

- Using fully connected layer assumes unimodality.
 - However, single phoneme has multiple allophones.
- Using softmax assumes in-distribution.
 - However, atypical speech is out-of-distribution.

MixGoP: Mixture distribution-based GoP

- Gaussian mixtures can model allophones.
- Softmax scores are replaced with Gaussian likelihood.

GoP vs. MixGoP



A vertical red decorative pattern on the left side of the slide, consisting of a dense grid of thin, slightly curved lines.

3. Experimental results

Datasets

- **Dysarthric datasets:** UASpeech (Kim et al., 2008), TORGO (Rudzicz et al., 2012), SSNCE (T. A. et al., 2016)
- **Non-native datasets:** speechocean762 (Zhang et al., 2021), L2-ARCTIC (Zhao et al., 2018)

Speech features

- **Traditional acoustic features:** MFCCs, Mel Spectrograms
- **TDNN-F features** (Zhang et al., 2021)
- **Self-supervised features:** XLS-R-300M (Babu et al., 2022), WavLM-Large (Chen et al., 2022)

Baselines

- **Phoneme classifier-based:** GMM-GoP (Witt et al., 2000), NN-GoP, DNN-GoP (Hu et al., 2015), MaxLogit-GoP (Yeo et al., 2023)
- **OOD detector-based approaches:** kNN (Sun et al., 2022), One-class SVM (Schölkopf et al., 2001), Phoneme-specific one-class SVM (Shahin et al., 2001), **MixGoP (Ours)**

Technical contributions

- Dysarthric + nonnative speech as a unified framework
- Extensive experiments (methods, datasets, etc.)
- <https://github.com/juice500ml/acoustic-units-for-ood>

Results

	GMM-G oP	NN-GoP	DNN-G oP	MaxLog it-GoP	kNN	oSVM	p-oSVM	Ours
UASpeech	0.568	0.568	0.548	0.558	0.606	0.537	0.327	0.623
TORGO	0.682	0.681	0.641	0.681	0.704	0.671	0.621	0.713
SSNCE	0.547	0.547	0.499	0.547	0.523	0.358	0.234	0.553
speechocean762	0.535	0.533	0.380	0.536	0.443	0.392	0.173	0.539
L2-ARCTIC	0.243	0.312	0.196	0.196	0.189	0.082	0.078	0.182

Ours show best performance across four out of five datasets!

Results

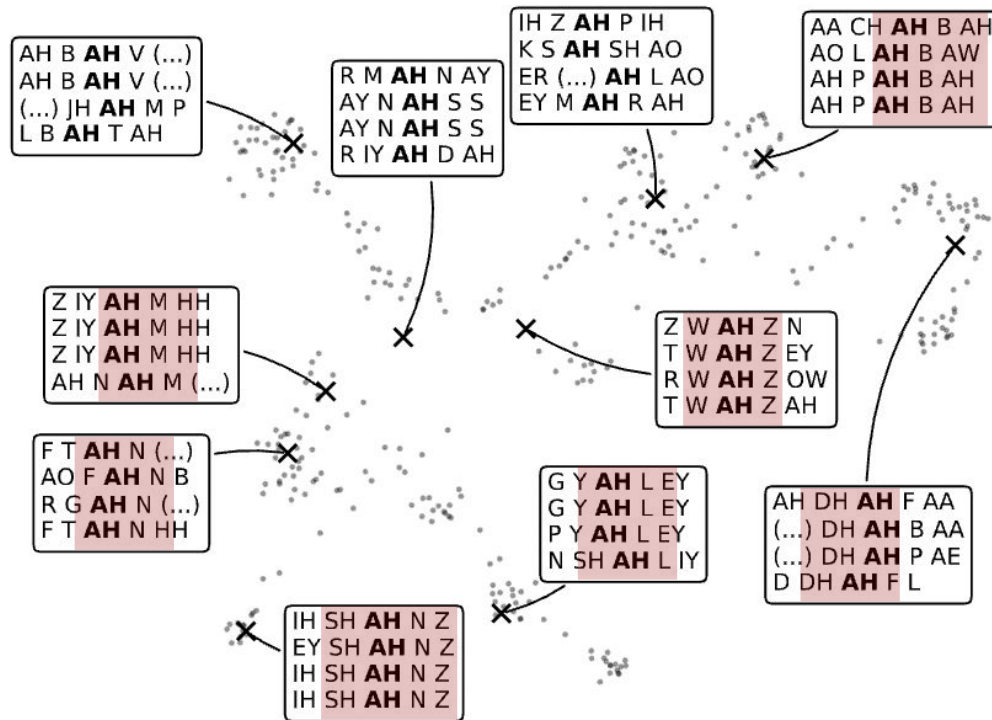
		GMM- GoP	MixGoP (Proposed)	
UASpeech	MFCC	0.428	0.182	Traditional features
	MelSpec	0.209	0.039	
	XLS-R	0.552	0.602	Self-supervised features
	WavLM	0.568	0.623	

- Compared to existing GoP, our MixGoP's performance improves a lot when using self-supervised features.
- Why is MixGoP more adapted to self-supervised features?

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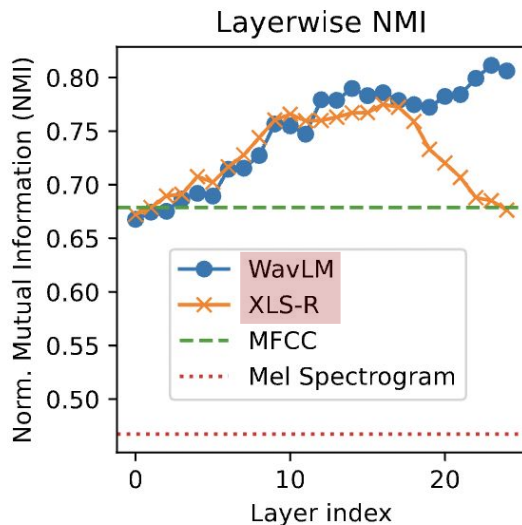
4. Allophony of self-supervised features

Observing self-supervised features

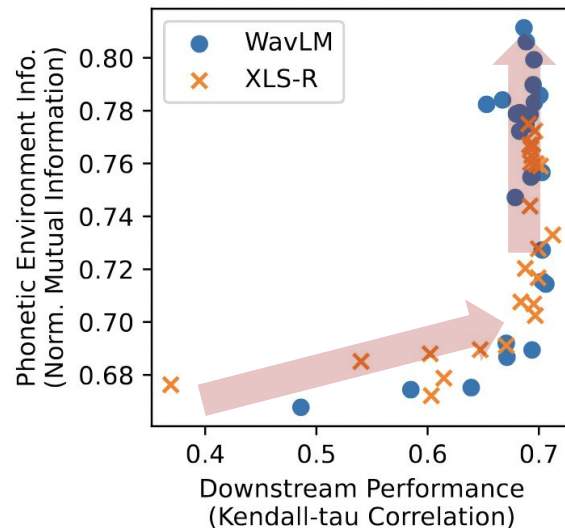


Self-supervised features are clustered based on their surrounding phonetic environments!

Measuring the allophony information



Self-supervised representations can capture allophony well.



Capturing allophony well leads to better assessment.

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5. Takeaways

Takeaways

- Self-supervised features cluster per allophony.
- Gaussian mixtures are useful to leverage such characteristics.
- We provide a unified framework of understanding atypical pronunciation assessment as an out-of-distribution task.

Thank you for listening!

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